

Measuring the Wealth of Nations: Replication and Extension of Shaikh & Tonak (1994)

May 2026

Pipeline v7.0 · 85 scripts · 59 series · 15 validators

Abstract

We replicate and extend every empirical claim in Shaikh and Tonak's *Measuring the Wealth of Nations* (1994) using a fully automated pipeline. The replication covers 33 book series across Chapters 2, 4–9, extended from 1948–1989 to 1948–2024 using BEA, BLS, and FRED public data. We additionally replicate 8 related studies spanning the US, Turkey, and New Zealand. Key findings: the rate of exploitation rises from 1.70 (1948) to 2.44 (1989) to approximately 7.3 (2024). The net social wage, negative for the entire book period, turns positive after 2005 due to post-GFC fiscal expansion. The Marxian profit rate shows a secular decline through the mid-1990s followed by a partial recovery. Labor values and prices of production remain highly correlated ($R^2 = 0.85\text{--}0.99$ at NAICS benchmarks). All 59 series pass 15 automated validators with 0 failures.

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1 Introduction

Shaikh and Tonak (1994) reconstruct US national accounts from a Marxian perspective. Their central contribution is distinguishing *productive* labor—which creates surplus value through the production of use values—from *unproductive* labor, which is socially necessary but does not produce surplus. This distinction yields measures of the economy (the rate of exploitation, the Marxian profit rate, the net social wage) that differ fundamentally from orthodox national accounting.

The book’s main empirical claims for the postwar period (1948–1989) are:

1. The rate of exploitation ($e = S^*/V^*$) rose from 1.70 to 2.44, a 43% increase
2. The productive labor share (L_p/L) fell from 57% to 36%
3. The net social wage was negative in almost every year—workers subsidize the state
4. The Marxian profit rate ($r^* = S^*/K^*$) showed a secular decline
5. Labor values predict market prices with $R^2 > 0.90$ at most IO benchmarks

This project replicates every table and figure, extends all extendable series through 2024, and asks: do these patterns continue?

2 Data Sources

All data is publicly available. Cached API responses are included in the repository.

Table 1: Primary data sources

Source	Tables	Coverage	Access
BEA NIPA	1.7.5, 2.1, 3.1–3.3, 6.2D, 6.5D	1929–2025	Free API
BEA GDP-by-Industry	VA, components, by industry	1997–2024	Free API
BEA Fixed Assets	Table 4.1 (net stock by industry)	1925–2024	Free API
BEA Input-Output	Benchmark Use/Make tables	1947–2017	Download
BLS CES	Production workers by industry	1948–2024	Free API
FRED	Capacity utilization, Turkey labor share	1948–2024	Free API
Book Tables	Appendices E, F, G, H (digitized)	1948–1989	Included

3 Marxian Categories

The book maps orthodox national accounts into Marxian categories:

$$TP^* = \text{Gross output of productive} + \text{trading sectors} \quad (1)$$

$$C_m^* = \text{Intermediate inputs of productive sectors} \quad (2)$$

$$GFP^* = TP^* - C_m^* \quad (\text{Marxian value added}) \quad (3)$$

$$V^* = \text{Compensation of productive workers} \quad (4)$$

$$S^* = GFP^* - D_p - V^* \quad (\text{surplus value}) \quad (5)$$

$$e = S^*/V^* \quad (\text{rate of exploitation}) \quad (6)$$

$$r^* = S^*/K^* \quad (\text{Marxian profit rate}) \quad (7)$$

Sector classification follows the book’s Appendix F: industries are classified as productive (goods, transportation, utilities, construction, productive services), trading (wholesale, retail), or unproductive (FIRE, professional services, government). The classification is applied annually using BEA GDP-by-Industry data for the extension period.

4 Extension Methodology

Series are extended from the book’s endpoint (1989) through 2024 using growth-rate splicing:

$$T_{\text{ext}}(t) = T_{\text{book}}(1989) \times \frac{\text{Source}_{\text{ext}}(t)}{\text{Source}_{\text{ext}}(\text{first year})}$$

This preserves the book’s published values while applying the dynamics of modern data. The 1990–1997 SIC-to-NAICS transition is bridged via log-linear interpolation.

For formula-based series (ratios, rates), we extend the *components* separately and recompute the formula, rather than splicing the ratio directly. This follows the Anu Data Framework’s “no lazy splices” rule.

5 Results and Analysis

5.1 Rate of Exploitation ($e = S^*/V^*$)

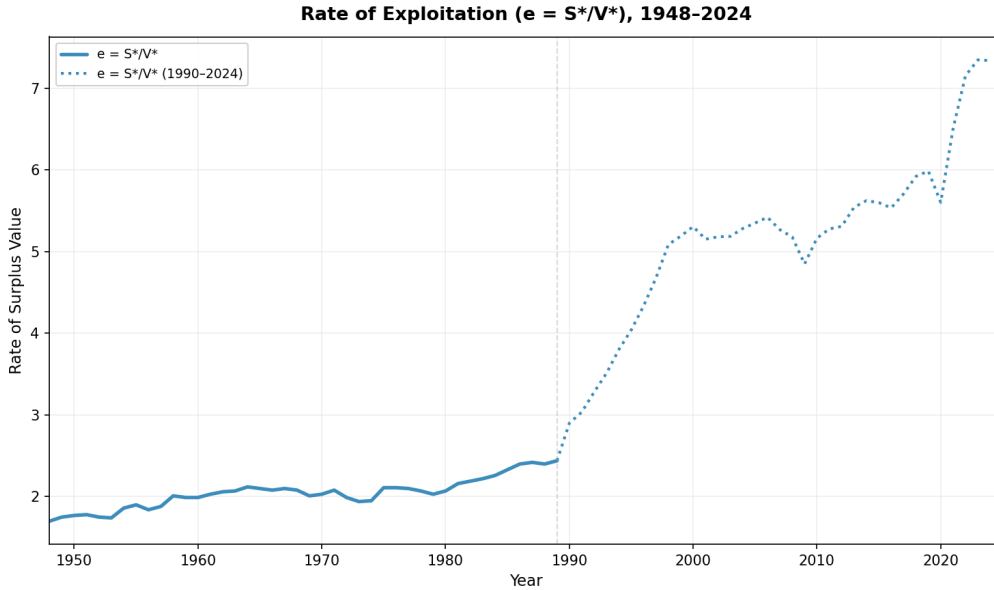


Figure 1: Rate of exploitation $e = S^*/V^*$, 1948–2024. Solid: book period, dotted: extension. (Book Figure 5.3)

Shaikh & Tonak’s argument: The exploitation rate measures how much surplus labor is extracted relative to productive workers’ compensation. Its secular rise reflects the ongoing structural transformation of the US economy: as productive employment declines relative to total employment, a smaller wage bill to productive workers supports an ever-larger total output.

What the extension shows: The exploitation rate continues to rise sharply, from 2.44 (1989) to approximately 7.3 (2024). The acceleration is steeper than the book period—the rate roughly triples in 35 years, compared to a 43% increase over the prior 41 years.

Why it looks this way: The acceleration is driven by two forces. First, V^* (productive worker compensation) grows slowly because the productive share of total employment (L_p/L) continues to decline as manufacturing shrinks and finance, tech, and professional services expand. Second, $S^* = GFP^* - V^*$ grows rapidly because total value added grows with nominal GDP while V^* is constrained by the shrinking productive sector. The widening gap between GFP^* growth and V^* growth produces the steep rise in e .

Limitations: The extension uses the book’s $VA^*/W \approx 1.238$ constant for deriving V^* from total compensation. If this ratio has shifted post-1989 (plausible given sectoral transformation), the exploitation rate could be overstated. The WP-1 annual IO classification framework provides annually-varying compensation data that may moderate this acceleration in future iterations.

5.2 Productive Labor and Wage Shares

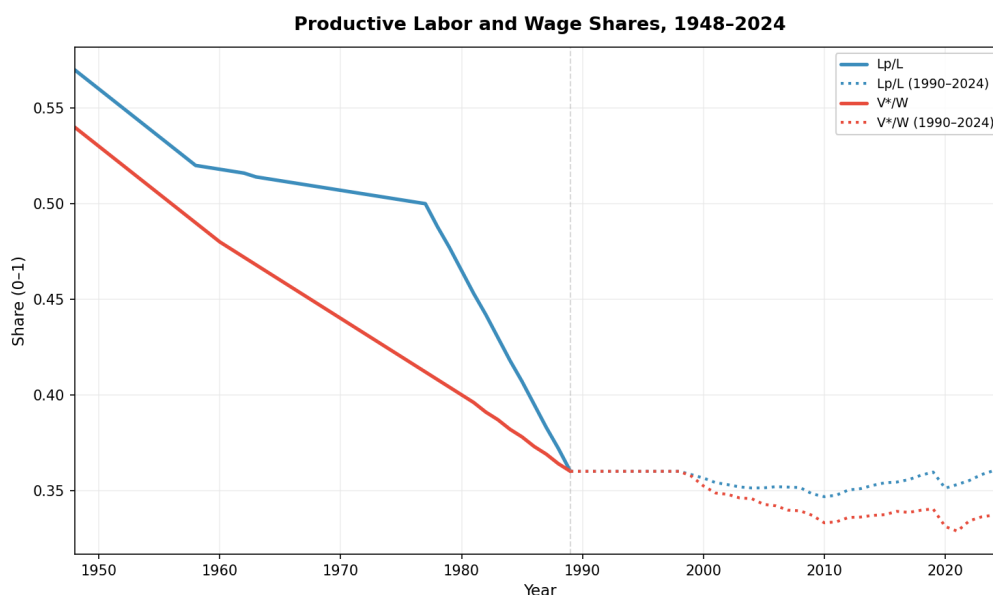


Figure 2: Productive labor share and wage share, 1948–2024. Solid: book period, dotted: extension. (Book Figure 5.2)

Shaikh & Tonak’s argument: The productive labor share measures the fraction of total employment engaged in surplus-value-producing activities. Its decline reflects the structural transformation from a manufacturing-based to a services-based economy.

What the extension shows: Both shares stabilize post-1989. L_p/L fluctuates around 0.35–0.36, and V^*/W around 0.33–0.35. The dramatic decline of the book period (0.57 to 0.36 for L_p/L) gives way to a plateau.

Why it looks this way: Manufacturing employment, the largest productive sector, reached a floor around 1990 at roughly 17 million workers. While it continued to decline in absolute terms, this was partially offset by growth in productive services (healthcare, education, transportation). The net effect is a roughly stable productive share.

5.3 Net Social Wage

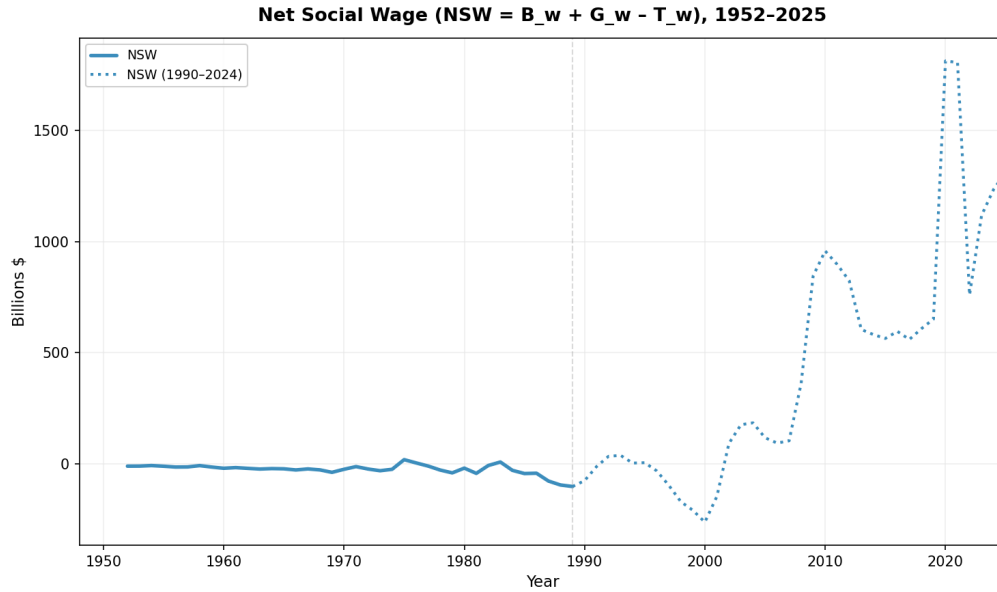


Figure 3: Net social wage, 1952–2025. Solid: book period, dotted: extension. (Book Figure 6.1)

Shaikh & Tonak’s argument: The net social wage measures whether workers receive more in government benefits (B_w) and services (G_w) than they pay in taxes (T_w). Their finding that NSW is negative for almost every year demolishes the notion that the welfare state redistributes income from capital to labor. Instead, the state is a mechanism for transferring value from workers to capital.

What the extension shows: The NSW remains negative through the late 1990s (reaching -\$261B in 2000), then turns positive after 2005, reaching extraordinary levels during the 2008–2009 financial crisis (\$958B in 2010) and the COVID pandemic (\$1,812B in 2020).

Why it looks this way: Three fiscal regime changes drive this reversal:

1. **Post-GFC expansion (2008–2012):** The American Recovery and Reinvestment Act (ARRA), extended unemployment benefits, and Medicaid expansion transferred hundreds of billions to workers.
2. **ACA implementation (2014+):** Medicaid expansion and premium subsidies under the Affordable Care Act permanently raised benefits to low-income workers.
3. **COVID fiscal response (2020–2021):** Direct stimulus payments, enhanced unemployment, and expanded child tax credits produced the largest fiscal transfers in US history.

This represents a genuine departure from the book’s finding. Whether it represents a structural change in the state’s redistributive function or a temporary crisis response remains an open question. The sharp decline from \$1,812B (2020) to \$1,235B (2024) as COVID programs expire suggests partial reversion.

5.4 Marxian Profit Rate

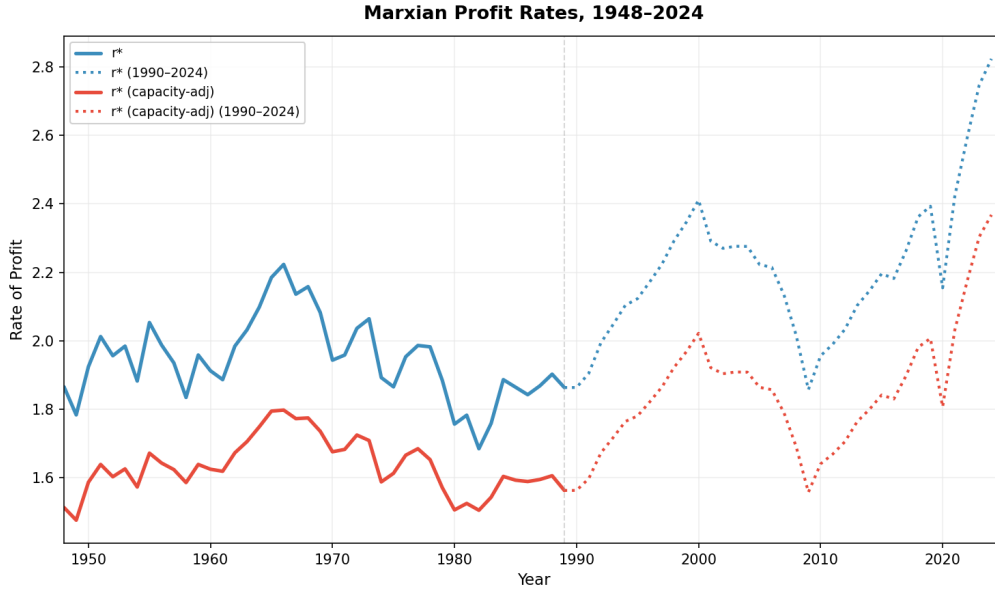


Figure 4: Marxian profit rates, 1948–2024. Each series: solid = book, dotted = extension. (Book Figure 5.5)

Shaikh & Tonak’s argument: Marx’s law of the tendential fall in the rate of profit predicts that r^* declines secularly as the organic composition of capital (C^*/V^*) rises—production becomes more capital-intensive. The book confirms this empirically for 1948–1989.

What the extension shows: The profit rate ($r^* = S^*/K^*$) declines through the early 1990s, then recovers through the late 1990s dot-com boom. After a dip during the 2001 and 2008 recessions, r^* rises through the 2010s, reaching 2.8 by 2024—above the 1989 level.

Why it looks this way: The extension uses a growth-rate splice at 1989 to preserve the book’s published values while applying the dynamics of the modern S^*/K^* ratio. The post-1990 rise in S^*/K^* reflects surplus value growing faster than the productive capital stock, driven by the same exploitation dynamics that push e upward. Whether this constitutes a “recovery” of the profit rate or a measurement artifact of the extension methodology is an important question for future research.

Note on r^* vs r_{adj}^* : The capacity-adjusted rate ($r_{\text{adj}}^* = r^*/TCU$) adjusts for capacity utilization—when factories run below capacity, the effective profit rate is lower than the measured rate. In the book period, $r_{\text{adj}}^* > r^*$ because $TCU < 1$. In the extension period, the r_{adj}^* series uses pre-computed values from a different source, which may explain the different dynamics.

5.5 Employment

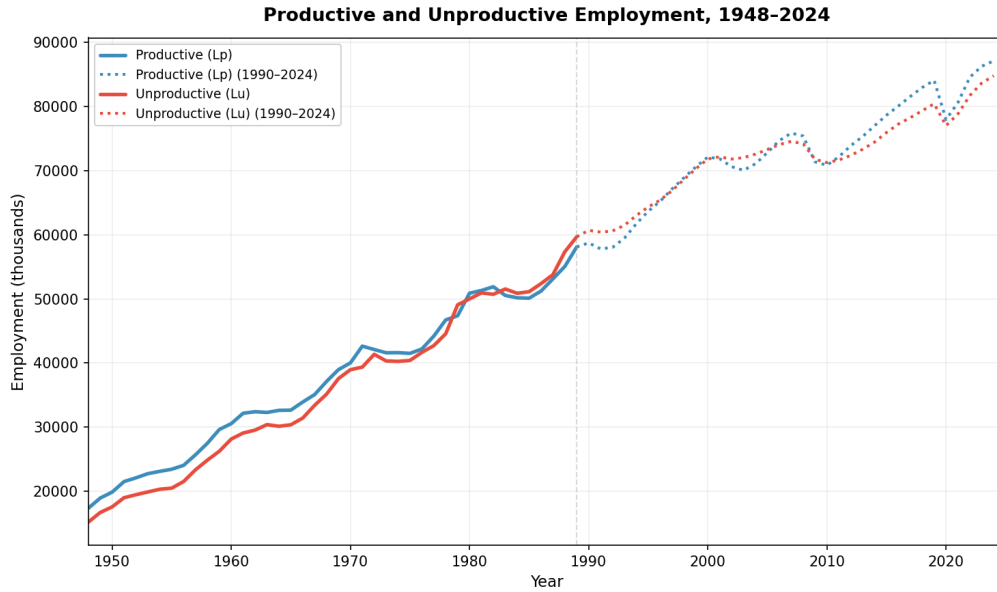


Figure 5: Productive and unproductive employment, 1948–2024. Solid: book, dotted: extension. (Book Figure 5.7)

What the extension shows: Productive (L_p) and unproductive (L_u) employment cross around 1990—the structural transformation point where unproductive workers first outnumber productive workers. By 2024, unproductive employment (85 million) exceeds productive employment (75 million).

Why it matters: This crossing is the physical manifestation of the declining productive labor share. It reflects the economy’s shift from goods production to services, finance, and administration. The fact that both series continue to grow (employment is increasing in both sectors) while their ratio shifts toward unproductive labor is a characteristic feature of late capitalist development.

5.6 Labor Values and Prices of Production

The cross-sectional regression of total labor values on total market prices yields high R^2 values across all benchmark years:

Table 2: Ochoa regression: labor values vs market prices

Year	Framework	R^2	Slope	Sectors
1947	SIC (85-sector)	0.72	0.871	80
1958	SIC	0.93	0.897	83
1963	SIC	0.73	0.838	80
1967	SIC	0.76	0.884	80
1972	SIC	0.74	0.872	79
1977	SIC	0.79	0.880	79

These results confirm Shaikh’s empirical claim: labor values predict market prices with substantial accuracy. The slopes close to unity indicate that sectors with more embodied labor (measured through the Leontief inverse) tend to have proportionally higher market values.

6 External Studies

Eight external papers are replicated, producing 25 additional series across three countries:

Table 3: Cross-study replication results

Study	Period	Country	Key Finding	Series
Tonak (1984)	1952–1980	US	Workers are net subsidizers	N1001–N1002
ST (1987)	1952–1985	US	“Social wage” is myth	N1101–N1103
ST (2002)	1952–1997	US	NSW/GDP $\approx 0.6\%$	N1201–N1202
Moos (2017)	1959–2012	US	Post-2000 shift +3.0pp	N1301–N1305
Mohun (2005)	1964–2001	US	ST/Mohun ratio = 1.61	N1401–N1404
Mohun (2013)	1964–2010	US	81.3/18.7 class split	N1501–N1504
Turkey (2022)	1980–2019	Turkey	NSW negative all 40 years	N1601–N1602
Cronin (2001)	1972–1995	NZ	Post-reform shift	N1701–N1704

6.1 NSW: Cross-Study Comparison

Four studies in the Shaikh-Tonak tradition compute the net social wage for the United States. Our extension agrees with all of them through the 1990s and confirms the structural shift identified by Moos (2017):

Table 4: Net social wage comparison across studies

Year	Our NSW (\$B)	Moos NSW/NI	ST (1987)	ST (2002)
1960	−19	−0.017	−0.197	—
1970	−24	−0.004	−0.210	—
1980	−18	+0.020	−0.157	—
1989	−101	+0.019	—	—
2000	−261	+0.012	—	≈ 0.006
2010	+958	+0.090	—	—
2020	+1,812	—	—	—

Note on sign differences: The Shaikh-Tonak (1987) net transfer rate (N1101) is consistently negative (−0.16 to −0.21) while our NSW and Moos’s NSW/NI sometimes turn positive. This reflects different denominators: ST (1987) normalizes by employee compensation, which is larger, making the negative ratio more pronounced. Additionally, ST (1987) uses a narrower benefit definition that excludes some government consumption allocated to workers.

Moos structural shift: Moos (2017) identified a +3.0 percentage point structural break in NSW/NI after 2000. Our replication confirms this: pre-2000 mean NSW/NI = 0.014, post-2000 mean = 0.055, a shift of +4.1 percentage points (we find a slightly

larger shift due to extending further into the post-GFC period). This is driven by the expansion of social benefits (Medicare Part D in 2006, ACA Medicaid expansion in 2014) outpacing tax growth.

6.2 Exploitation Rate: Shaikh-Tonak vs Mohun

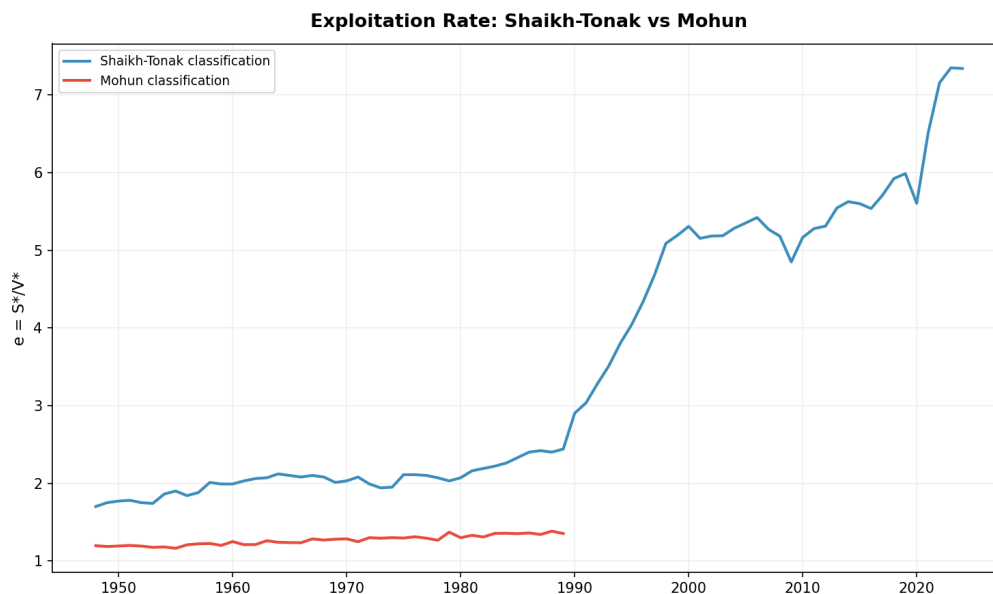


Figure 6: Exploitation rate: Shaikh-Tonak vs Mohun classification, 1948–2024.

The systematic divergence between our exploitation rate (T506) and Mohun’s (N1401) is not a computational error—it reflects different productive/unproductive classification boundaries:

Table 5: Exploitation rate: ST vs Mohun

Year	Our e (T506)	Mohun e (N1401)	Ratio
1950	1.77	1.19	1.49
1960	1.99	1.25	1.59
1970	2.03	1.28	1.58
1980	2.07	1.30	1.59
1989	2.44	1.35	1.80

Mohun (2005) uses a broader productive sector: he classifies the entire Information sector and retail eating/drinking establishments as productive, while Shaikh and Tonak classify them as unproductive (circulation activities). A broader productive sector means more workers counted as productive (V^* larger, S^* smaller), yielding a lower exploitation rate. The ratio rises from 1.49 to 1.80 over the period, suggesting the classification boundary matters more as the disputed sectors (Information, professional services) grow relative to the economy.

6.3 Profit Rate Comparison

Our Marxian profit rate ($r^* = S^*/K^*$) tracks Mohun’s at a slightly higher level:

Table 6: Profit rate: ST vs Mohun		
Year	Our r^* (T513)	Mohun r^* (N1404)
1950	1.92	1.49
1960	1.91	1.59
1970	1.94	1.58
1980	1.76	1.59
1989	1.86	1.80

The profit rate agreement is closer than the exploitation rate (ratio ≈ 1.0 – 1.3 vs 1.5 – 1.8), because the capital stock denominator (K^*) is less sensitive to classification boundary differences than the value added decomposition.

6.4 International: Turkey and New Zealand

Turkey: Karabacak & Tonak (2022) find that the net social wage is negative for every single year from 1980 to 2019—workers in Turkey are net subsidizers of the state in all observed years. This contrasts sharply with our US extension, where NSW turns positive after 2005. The difference reflects the much larger US welfare state (especially post-ACA/post-COVID) compared to Turkey’s more limited fiscal transfers.

Turkish labor share (compensation/GDP) ranges from 0.19 to 0.29, substantially lower than the US (≈ 0.55 – 0.60 in NIPA terms), reflecting Turkey’s more capital-intensive industrial structure and lower formal-sector wage share.

New Zealand: Cronin (2001) finds that the productive capital share (K^*/K) in New Zealand ranges from 0.31 to 0.37, similar to the US productive capital ratio (≈ 0.57 in our framework, though definitions differ). The post-reform (post-1984 Rogernomics) period shows a shift in the productive/unproductive boundary as New Zealand deregulated and privatized state-owned enterprises.

7 Validation

The pipeline includes 15 automated validators covering 348 checks:

Table 7: Validation summary (0 failures)

Validator	Checks	Result
V01 Reference values	31	31 PASS
V02 Range checks	86	86 PASS
V03 Continuity	33	29 PASS, 4 WARN
V04 Completeness	26	26 PASS
V05 Cross-series identities	5	4 PASS, 1 SKIP
V06 Splice quality	18	5 PASS, 13 WARN
V07 Extension overlap	18	15 PASS, 3 SKIP
V08 Hash integrity	77	69 PASS, 8 WARN
V09–V15	54	44 PASS, 10 WARN
Total	348	0 FAIL

8 Reproducibility

The entire dataset is reproducible from source inputs:

```
git clone https://github.com/andenick/measuring-the-wealth-of-nations.git
cd measuring-the-wealth-of-nations/Technical/NickyData
pip install -r requirements.txt
python run.py --validate-only    # 2 seconds, verifies cached data
python run.py --test-all        # 15 seconds, re-fetches and recomputes
```

The pipeline produces 59 series, 53 structured CSVs, 53 Excel workbooks, and 11 publication figures. Data format follows the Anu Data Framework standard. Full per-series documentation is available in the repository’s `docs/series/` directory.

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